
A Robot Policy Designed as a Weight-Space Artifact

Exact Tensor Structure in a Capable VLA

Rational conversion, reduced-Gram analysis, and causal intervention

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Abstract

Tensor decomposition rewrites a multidimensional table as a small set of low-rank factors. In a neural network, those factors can expose which combinations of inputs and outputs are encoded by the weights. Conventional nonlinear operations make such a table difficult to obtain directly. We introduce χ -VLA, a 20.1M-parameter vision-language-action policy containing only bilinear, linear, and rational learned operations. Its checkpoint therefore specifies an explicit rational tensor program rather than only an opaque executable function. A mechanical audit and local reconstructions verify that claim. We derive an exact weight-based decomposition of each softmax-free attention layer and reconstruct all 12 modules in three checkpoints with worst relative error 2.56×10^{-7} . An $O(d^2)$ reduced Gram avoids materializing the coefficient tensor. All 36 primary block-head tests favor the rank-128 weight-derived subspace over fixed random controls. The full appendix screen finds the effect in 83/96 pairs and identifies where the signal concentrates across depth. In a complementary exploratory rank screen on the three ViT checkpoints, a causally ranked 96-dimensional visual subspace succeeds in 50/60 matched trials, compared with 52/60 for the full representation. Three equal-rank random controls reach only 1/60, 3/60, and 3/60. The selected subspace retains 49 of the 52 full-policy successes. On disjoint cache samples, it has $33.5\text{--}110.6\times$ lower reconstruction error than the median across three random projectors, computed separately by action group. A convolutional instantiation of the same rational model family reaches $93.7\% \pm 0.8\%$ on LIBERO-Object, and its three-checkpoint ensemble reaches 96.2%. We therefore establish a capable model family with exact layerwise structure and a complementary compact exploratory policy bottleneck. We do not yet map the exact attention factors directly to that causal basis.

1 Introduction

Neural checkpoints are usually treated as opaque containers for functions. We can run them and collect their activations, but the weights alone rarely provide a readable map from input features to outputs. A post-hoc probe may reveal that an activation correlates with behavior, yet it does not show how the weights construct that feature or whether editing it will have a selective causal effect.

A tensor is a multidimensional table, and tensor decomposition is the analogue of low-rank matrix factorization for such tables. When a network uses only additions, multiplications, and ratios of polynomials, its weights can be expanded into coefficient tensors whose entries record input interactions. Decomposition ranks simpler factors inside those coefficients. The intended benefit is an inspectable model artifact: candidate mechanisms come from the trained parameters, then activation data and interventions test whether they matter. Softmax, ordinary activations, and square-root

normalization obstruct this direct conversion in conventional networks. Bilinear language and vision models suggest that changing the primitives can recover it [3, 4]. Whether the resulting artifact can still control a robot is unresolved.

We study two linked questions:

Can the learned weights of a specialist VLA specify an explicit rational tensor program, and can exact weight analysis and causal representation tests coexist in that model family?

Our contributions are:

1. We construct and mechanically certify a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We derive exact weight-based structure through each softmax-free attention layer and an $O(d^2)$ reduced-Gram calculation that avoids materializing its high-order coefficient tensor.
3. We test two complementary analysis surfaces. Exact attention subspaces beat fixed random controls. Separately, an exploratory causally ranked rank-96 visual subspace retains 49 of 52 full-policy successes across three checkpoints. Three random controls total only 7/180.
4. We establish capability in a convolutional instantiation of the same rational family, reaching $93.7\% \pm 0.8\%$ over three LIBERO-Object seeds. A three-checkpoint ensemble reaches 96.2% at three times the inference cost.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes exact weight structure directly accessible and supports separate causal tests of its internal representations.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive replaces a familiar transformer component while keeping explicit coefficients. Linear maps contribute first-order terms, while multiplying learned projections creates higher-order interactions that remain determined by the weights.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

The inner sum is a table indexed by one output and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization.

Softmax-free bilinear attention. Each head forms two query-key systems and one value stream:

$$Y = C[M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations, with interaction coefficients determined by the weights.

Rational per-instance normalization. RMS normalization contains an input-dependent square root. We instead deploy a fixed rational approximation $r(z) = P(z)/Q(z)$ to $1/\sqrt{z + \epsilon}$:

$$\text{RNorm}(x) = x r \left(\frac{1}{d} \sum_{j=1}^d x_j^2 \right). \quad (3)$$

Here rational means a ratio of polynomials. Projective coordinates carry each activation as a numerator-denominator pair (N, D) representing N/D . Updating this pair preserves input-specific rescaling without leaving the rational tensor representation.

2.2 Architecture and training

The policy follows the high-level VLA sequence of visual encoding, language and state embedding, joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven continuous action dimensions. The complete model has 20,137,352 learned parameters.

Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay, bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks that verify every requested state was loaded.

2.3 Mechanical certificate on the trained checkpoint

Architecture definitions can differ from deployed code. We therefore audit the real 189/200 checkpoint at runtime and reconstruct representative operations from saved weights.

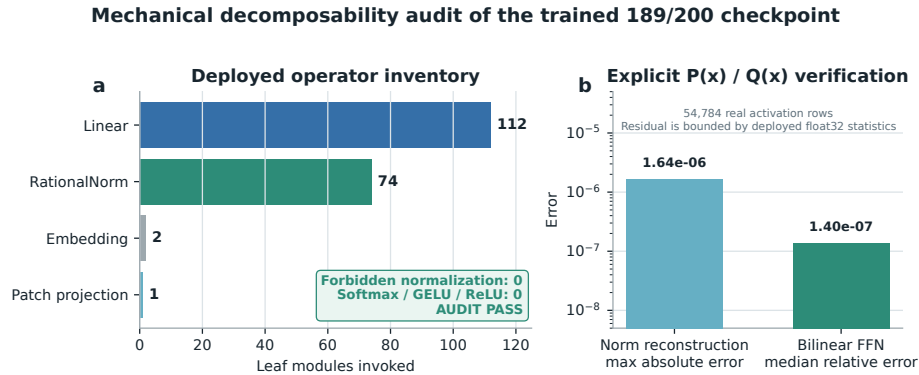


Figure 1: **Mechanical audit of the trained rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.

The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN reconstruction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed module’s float32 statistic. The audit certifies operator membership and local reconstruction. It does not materialize one compact polynomial for the full eight-layer transformer.

3 The artifact is a capable closed-loop policy

3.1 Evaluation protocol

We evaluate all ten LIBERO-Object tasks with:

- 50 canonical trials per task
- a 280-step cap
- three independently trained seeds per architecture
- 500 rollouts per seed and 1,500 rollouts per architecture

Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not rerun in our codebase, so differences may still include implementation and training choices.

3.2 Results

Method	Parameters	Training seeds	Success
χ -VLA, rational conv	20.1M	3	93.7% $\pm$ 0.8%
χ -VLA, conv ensemble	$3 \times 20.1\text{M}$	$3 \rightarrow 1$	96.2%
OpenVLA-7B [1]	7B	3	88.4% $\pm$ 0.8%
Diffusion Policy [1]	not matched	3	92.5% $\pm$ 0.7%

Table 1: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

The convolutional seeds score 94.0%, 94.4%, and 92.8%. Their narrow spread shows that high specialist capability is reproduced across three independently trained checkpoints.

Elementwise averaging of the three convolutional action predictions reaches 96.2%. This is 2.3 points above the mean of its members and 1.2 points above the strongest member. The gain is largest on tomato sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. It is an existence and reproducibility claim for a compact tensor-decomposable policy, not a claim of broad robotic generalization or state of the art.

4 Exact weight-derived structure through attention

4.1 Exact layerwise construction

Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives a coefficient table whose entries measure how combinations of query, key, and value coordinates contribute to an output. No examples or fitted probe are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

The deployed attention also normalizes each query and key with Equation 3. Each normalization multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled outside the bilinear dot products. The polynomial attention core is unchanged, while explicit numerator and denominator factors carry the normalization. The rational extension agrees to 3.61×10^{-16} . We then reconstruct all four vision and eight joint attention modules in each of three trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the worst absolute difference is 2.23×10^{-5} . This residual comes from the deployed implementation’s intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15} values. We instead form an $O(d^2)$ reduced Gram. Like a covariance matrix, its leading eigenvectors rank important directions without constructing the full table. It matches brute-force materialization to 2.13×10^{-14} at toy scale.

4.2 Trained policy result

For the subspace-ranking experiment, we apply the exact weight-derived calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are sampled once, then fixed and reused for all examples. The weights choose the subspace, while cached activations are used only afterward to score how well it preserves the layer’s computation.

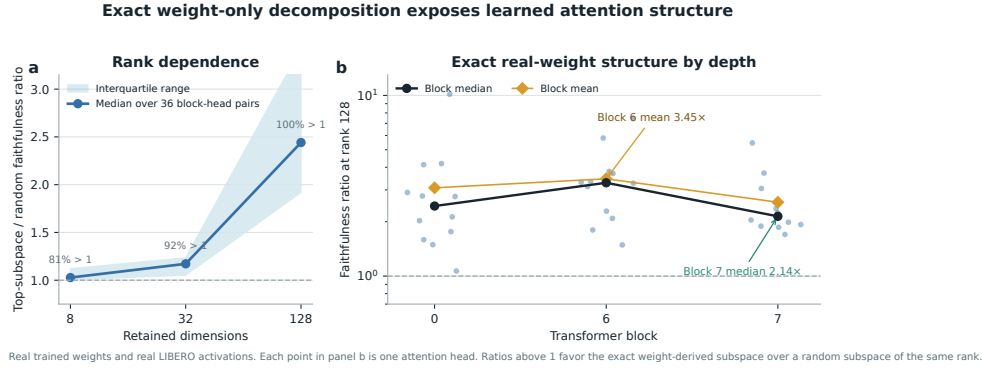


Figure 2: Exact weight-derived attention structure. Ratios above one favor the exact weight-derived subspace over a random equal-rank subspace. Each point in the right panel is one trained attention head.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks 0, 6, and 7.

This extends exact weight-only construction through individual attention layers. It is not an exact end-to-end decomposition of the complete policy. Composition across all eight blocks still faces rapid degree and representation growth.

5 Causal policy structure

We next test the policy’s representation with a separate causal analysis.

5.1 Causal visual subspace

At the visual bond before the joint backbone, we build one shared action-balanced ranking. We form one Jacobian Gram per action coordinate, trace-normalize the seven Grams, and average them. A subspace is a smaller coordinate system inside the model’s 384-dimensional visual representation. Retaining it means zeroing all other directions, then running the actual policy. We compare selected and random equal-width subspaces. Because the ranking uses cached training demonstrations and downstream action gradients, offline reconstruction remains in-cache. Its evaluation samples are disjoint from the samples that estimate the ranking. Closed-loop intervention uses independent canonical simulator states, so it tests a causal bottleneck but is not a weight-only discovery result.

The exploratory action-balanced screen tests ranks 64, 96, 128, and 192 on tasks 4 to 7. Appendix Figure 3 reports the full curve. Rank 96 is the smallest tested width with more than 90% conditional retention while random success remains below 5%. Tasks 8 and 9 are reserved for a frozen confirmation and are not used here. At rank 96, full representations succeed in 17/20, 15/20, and 20/20 trials across checkpoints. Selected subspaces reach 17/20, 14/20, and 19/20, and retain 49 of the 52 exact full-policy successes. Their conditional retention is 94.1%, 93.3%, and 95.0%. Three random controls per checkpoint total only 1/60, 3/60, and 3/60. Every checkpoint and random-control comparison has the same selected-over-random direction. Within each fixed checkpoint, the descriptive exact paired tests have $p < 2.5 \times 10^{-4}$. Using 1,024 discovery samples and 1,024 disjoint evaluation samples per checkpoint, the selected basis has 53.0–110.6× lower translation

error, $37.5\text{--}62.1\times$ lower rotation error, and $33.5\text{--}103.0\times$ lower gripper error than the median across three random projectors, computed separately by action group. This repairs sample reuse but does not turn the training-cache diagnostic into held-out task generalization. This data-driven full-policy ranking complements the exact layerwise analysis in Section 4.

6 Related work and limitations

Related work. OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control baselines [1, 2]. Bilinear MLPs, χ -nets, and bilinear IMPALA policies show that multiplicative weights can expose low-rank structure and causal controls [3–5]. Recent work also steers pretrained VLAs through activation directions [6]. Our distinction is the exact weight-derived construction in a policy designed as a rational tensor program. Tensor and polynomial networks provide the broader multilinear foundation [12–14]. VLA robustness benchmarks show that standard LIBERO success can coexist with weak language use [9–11].

Limitations and conclusion. The benchmark is one specialist suite, so it does not establish broad robotic generalization. Exact reconstruction is certified layer by layer, while the full-policy visual ranking also uses cached activations and action gradients. The capability result uses convolutional checkpoints, while the exact and causal analyses use separate ViT checkpoints. The latter are complementary, not a demonstrated direct mapping. Evaluation uses fixed task instructions and does not establish robust instruction grounding. Within that scope, rational restrictions support strong LIBERO-Object control, exact trained-weight certificates, and an exploratory rank-96 causal bottleneck that retains 49 of 52 full-policy successes across three checkpoints. We do not claim a direct coefficient-edit interface.

Reproducibility. The final artifact will provide code, configurations, checkpoint hashes, exact certificates, and per-episode logs. Canonical-state checks fail loudly when requested states are unavailable.

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A Supporting results and decisions

A.1 Rank screen and selectivity

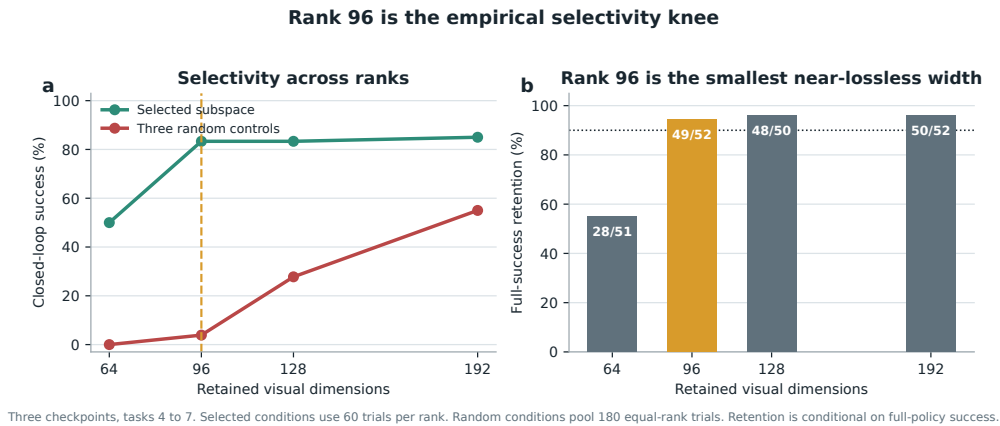


Figure 3: **Rank 96 is the empirical selectivity knee.** Selected subspaces use 60 closed-loop trials per rank across three checkpoints. Random results pool three equal-rank controls per checkpoint, for 180 trials per rank. Each rank has its own paired full-policy condition. Rank 96 is the smallest width above 90% conditional retention, while random success remains below 5%.

218 The selected subspace rises from 30/60 successes at rank 64 to 50/60 at rank 96. Wider ranks
 219 add little selected success, reaching 50/60 and 51/60 at ranks 128 and 192. In contrast, random
 220 success rises from 7/180 at rank 96 to 50/180 and 99/180. Conditional retention is 54.9%, 94.2%,
 221 96.0%, and 96.2% across the four ranks. Thus rank 96 preserves nearly all baseline successes before
 222 equal-width random subspaces become broadly capable.

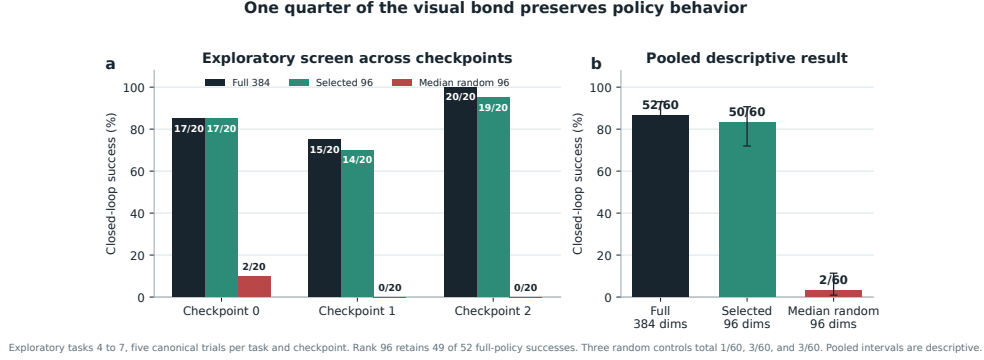


Figure 4: **One quarter of the visual bond preserves policy behavior in the exploratory rank screen.** Each checkpoint uses four tasks and five paired canonical trials per task. Rank 96 retains 49 of 52 full-policy successes. Three equal-rank random controls reach 1/60, 3/60, and 3/60. The pooled interval is descriptive because trials within a checkpoint share learned weights.

223 A.2 Per-task ensemble behavior

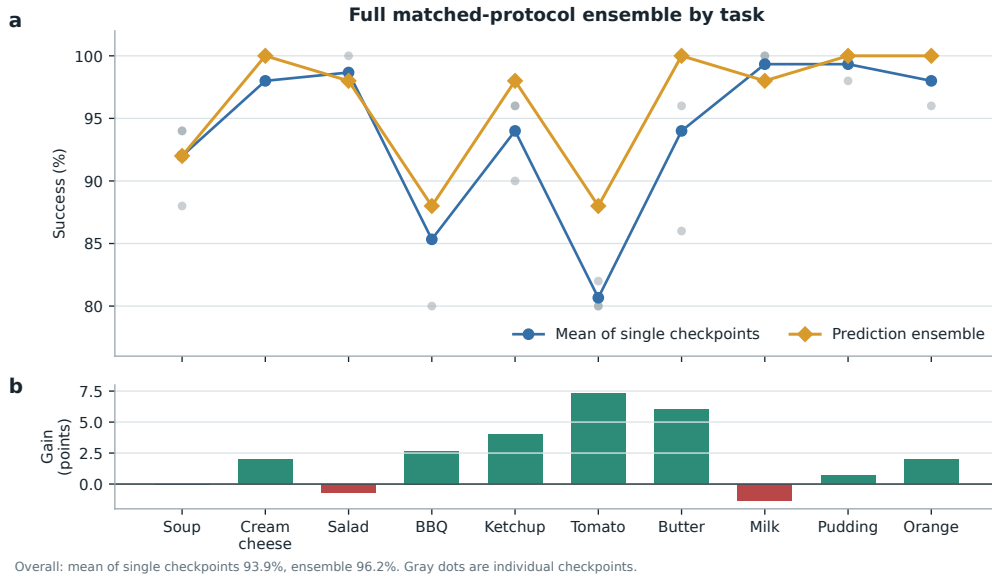


Figure 5: **Per-task effect of prediction averaging.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 20.1M-parameter policy.

224 A.3 Expanded attention screen

225 The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128,
 226 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean

227 is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only 54/96 and 60/96
228 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider
229 tested rank.

230 **A.4 Numerical scope of rational conversion**

231 The deployed Padé computation is exactly rational even though it approximates RMS normalization.
232 Final certification must report its polynomial degrees, coefficients, denominator margin, observed
233 activation range, tail error, action drift, and runtime overhead. Operator membership alone does not
234 establish practical numerical safety under distribution shift.